

Puspha Raj Pandeya

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Education

Southern Illinois University Edwardsville

Master of Science in Computer Science. GPA 3.455 (Cumulative after 2 semesters).

Relevant Coursework: AI, Computer Vision, Machine Learning, and Data Mining.

Edwardsville, IL

Aug 2024 - May 2026

Nepal College of Information Technology

Bachelor of Computer Engineering. GPA 3.03.

Relevant Coursework: AI, Data Mining, NLP, Big Data.

Lalitpur, Nepal

Sept 2017 - Aug 2022

Skills & Interests

Software Development: Python · Pandas · SQL · MongoDB · Git · Docker · Flask · FastAPI · HTML · CSS

Data: Data Engineering · Machine Learning · Computer Vision · NLP · ETL · ELK · Big Data · Airflow · Kafka

General Tools: Microsoft Word · Excel · PowerPoint · Power BI · DBeaver · MySQL Workbench · AWS

Soft Skills: Time Management · Organizational Skills · Interpersonal Communication

Interests: Chess and Music.

Experience

Cedar Gate Technologies

Data Engineer

Lalitpur, Nepal

Jun 2023 – Aug 2024

- Analyzed US healthcare data to identify patient cost distributions and the legitimacy of insurance claims, contributing to the analysis of performance drivers and methods to control cost, quality, and risk within value-based care programs.
- Cleaned, transformed, aggregated, and mapped raw patient data to a company-wide schema using a proprietary data management system to onboard data from disparate sources. This work helped to normalize, augment, and enrich data into a single data lake serving as the source of truth.
- Standardized key fields (e.g., relationship codes, insurance plans) and performed member information matching across data sources to enable the use of fully enriched member data for analytics. Utilized statistical techniques to interpret and analyze member demography and member months data, supporting functions like assessing performance and simplifying member attribution.

Dekods INC.

Freelance Data Analytics Consultant

Kathmandu, Nepal

Jan 2023 – Feb 2023

- Cleaned and processed 100,000 record files for a UNICEF initiative to upgrade various infrastructures in Nepal's rural communities.
- Performed exploratory data analysis using Python and Pandas, and data cleaning and transformation scripts were created.
- Created and implemented a Power BI dashboard for the UNICEF project team to display important metrics.
- Used DAX and best practices in visualization to deliver insights into rural families.

- Developed sales forecasting models using time series analysis techniques like ARIMA and SARIMA.
- Created a recommendation system using content and collaborative filtering methods, using cosine similarities and Non-negative matrix factorization techniques.
- Developed a chatbot for an e-commerce client using NLP and neural networks.
- Contributed to developing a Logistics Management System to improve product delivery by finding the shortest path and using resources efficiently. Dijkstra's algorithm was used to find the shortest path, and a navigation system was built using OpenStreetMap.
- Designed and implemented a data pipeline using the ELK stack and Python to create a sales analysis dashboard in Kibana.
- Designed and implemented efficient ETL processes to integrate data from multiple databases into comprehensive data warehouses, enabling faster and more accurate data analysis and visualizations.
- Designed, built, and containerized scalable backend systems for multiple projects using Flask, FastAPI, and MongoDB, resulting in efficient and easily maintainable applications.
- Migrated data from SQL to NoSQL databases using SQL and Pandas for denormalization and cleaning.

Certifications

AWS Academy Graduate - AWS Academy Data Analytics 
Python for Everybody Specialization 
Capstone: Retrieving, Processing, and Visualizing Data with Python 
Python Data Structures 
Using Databases with Python 

Amazon Web Services
Coursera
Coursera
Coursera
Coursera

Projects

Brain Tumor Segmentation

SIUE, Computer Vision, CS 590

- Performed automated segmentation of lower-grade gliomas in MRI scans by comparing deep learning models, achieving a Dice Coefficient of 64.83% and IoU of 62.94% with U-Net, which outperformed SegNet in accuracy. This analysis utilized a curated dataset of 110 patients and 3,929 MRI slices from TCGA/TCIA.
- Analyzed model performance across frameworks and demographic subgroups, confirming U-Net's superior segmentation accuracy and consistency across genders with Dice Coefficients of approximately 64.24% for males and 64.94% for females. The project addressed the critical challenge of automating brain tumor segmentation to improve diagnostic precision and reliability.

Sentiment Analysis for Business Reviews

NCIT, Project III, CMP 490

- Developed and evaluated a sentiment analysis system for business reviews to automate the detection of positive, negative, and neutral sentiments, comparing Transformer-based models (like BERT) against traditional Neural Networks (like CNN).
- Analyzed model performance across frameworks, achieving approximately 92% accuracy with BERT compared to 85% accuracy with CNN in sentiment classification. The project utilized a dataset comprising millions of business reviews to address challenges in automated sentiment detection.